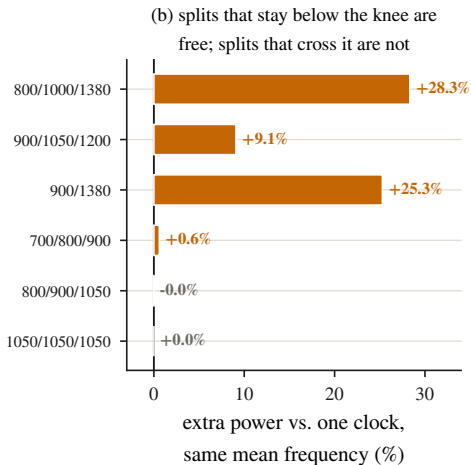
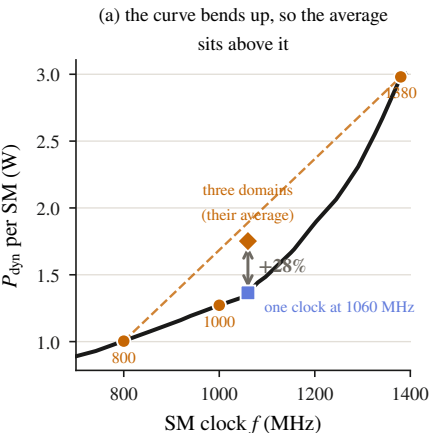




Same work, same time, same 108 SMs  $\Rightarrow$  the SM-weighted mean frequency is fixed. Energy is  $T \sum_d g(f_d)$  with  $g(f) = f \kappa(f)$ , so by Jensen a split can only beat the uniform clock if  $g$  is concave. Measured,  $g$ 's slope rises 1.23  $\rightarrow$  8.02 mW/MHz/SM: convex.  $\kappa$  from data/kappa\_sweep.csv, per SM.